

WORKSHOP HANDOUT

Data Analysis for Public Health Using R

A Complete 5-Day Training Programme

Dataset	Records	Key Variables	Used For
infant.xls	200	id, age, sex, wt1, ht1, water, maeduc, wt2	Days 1–5 (infant growth analysis)
lbw.xlsx	189	ID, low, age, smoke, race, ht, ui, lwt, bwt	Days 1–5 (low birth weight study)

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How to Use This Handout

This handout is your complete reference guide for the 5-Day R Workshop. It contains all the R code from the training sessions, step-by-step explanations, tasks with answers, and quick-reference tables you can use after the workshop.

This handout contains:

- All R code from Days 1–5, ready to copy and run
- Explanation of every function and concept covered
- 4 hands-on tasks per day (20 tasks total) with guidance
- Quick-reference tables for common functions
- Variable dictionary for both infant and lbw datasets
- Next steps and recommended resources

Software Requirements

Before running any code in this handout, make sure you have the following installed:

- R (latest version) — download from: <https://cran.r-project.org>
- RStudio (free Desktop edition) — download from: <https://posit.co/downloads>
- The two datasets: infant.xls and lbw.xlsx — place in your working directory

```
# Install all required packages (run once only, then remove the # next time)
install.packages('readxl')      # Read Excel files
install.packages('tidyverse')   # Data wrangling + ggplot2 + more
install.packages('broom')       # Tidy model output
install.packages('writexl')     # Export to Excel
```

Dataset Variable Dictionary

Dataset 1: lbw.xlsx — Low Birth Weight Study (n = 189)

This dataset comes from a study investigating risk factors for low birth weight (birth weight < 2,500 grams). Each row is one birth record.

Variable	Type	Values / Range	Description
ID	Integer	1 – 189	Unique identifier for each birth
low	Binary	0 = Normal, 1 = Low BW	Low birth weight indicator (outcome variable)
age	Numeric	14 – 45 years	Mother's age in years
lwt	Numeric	80 – 250 lbs	Mother's weight at last menstrual period
race	Categorical	1=White, 2=Black, 3=Other	Mother's race — convert to factor
smoke	Binary	0 = No, 1 = Yes	Smoking status during pregnancy
ptl	Integer	0, 1, 2, 3	Number of premature labors
ht	Binary	0 = No, 1 = Yes	History of hypertension
ui	Binary	0 = No, 1 = Yes	Presence of uterine irritability
ftv	Integer	0, 1, 2, 3, 6	Number of first trimester physician visits
bwt	Numeric	709 – 4990 g	Birth weight in grams (continuous outcome)

Dataset 2: infant.xls — Infant Growth Study (n = 200)

This dataset contains growth measurements from 200 infants. Each row is one infant with two weight measurements (baseline and follow-up).

Variable	Type	Values / Range	Description
id	Integer	1 – 200	Unique identifier for each infant
age	Numeric	1 – 60 months	Infant age in months
sex	Binary	0 = Male, 1 = Female	Sex of the infant — convert to factor
wt1	Numeric	Varies	Baseline weight (first measurement)
ht1	Numeric	Varies (cm)	Height at baseline in centimetres
water	Binary	0 = No, 1 = Yes	Access to clean drinking water
maeduc	Integer	0, 1, 2, 3, 4	Mother's education level (coded)
smeg	Numeric	0 – 600+	Smegma score (clinical measurement)
wt2	Numeric	Varies	Follow-up weight (second measurement)

Derived Variables (you will create these in the sessions)

- $wt_gain = wt2 - wt1$ — weight gained between measurements
- sex_factor — factor version of sex (Male / Female)
- $smoke_factor$ — factor version of smoke (Non-Smoker / Smoker)
- $race_factor$ — factor version of race (White / Black / Other)
- $bwt_kg = bwt / 1000$ — birth weight in kilograms
- $lbw_category = 'Low\ BW'$ if $bwt < 2500$, else $'Normal\ BW'$
- $gained_weight = 1$ if $wt2 > wt1$, else 0 (binary outcome for Day 5)

DAY 1

Getting Started with R & RStudio

Arithmetic · Objects · Help System · Packages · Loading Data

Session I: Why R for Public Health?

Session II: RStudio Tour & Basic Arithmetic

Session III: The Help System

Session IV: Packages & Loading Data

Day 1: Getting Started with R & RStudio

Today you will take your first steps in R. By the end of the day you will be able to load your two public health datasets into R and perform basic operations on them.

Session I — Why R for Public Health?

This session is a facilitated discussion. Key points to remember:

- R is completely FREE — no license cost unlike SPSS (~\$1,000+/year) or STATA
- Reproducible: your script records every step — anyone can re-run your analysis exactly
- All-in-one: data cleaning, statistics, AND publication-quality graphics in one place
- Globally trusted: used by WHO, CDC, NIH, ECDC, and thousands of researchers
- Massive community: millions of users and hundreds of free online tutorials

Session II — RStudio Tour & Basic Arithmetic

The Four RStudio Panels

Panel	Location	Purpose
Script Editor	Top-Left	Write and save your code here. Run a line with Ctrl+Enter (Cmd+Enter on Mac)
Console	Bottom-Left	Where results appear. Type directly here for quick one-off commands
Environment	Top-Right	Shows all objects (datasets, variables) currently in R's memory
Files/Plots/Help	Bottom-Right	Browse files, view plots, and read function documentation

R as a Calculator

```
# Basic arithmetic
100 + 250          # Addition → 350
1000 - 375         # Subtraction → 625
60 * 7             # Multiplication → 420
189 / 3            # Division → 63
2^8               # Power → 256
sqrt(144)          # Square root → 12
(10 + 20) * 3      # Brackets are evaluated first → 90

# Creating objects (variables) with the assignment arrow <-
total_births      <- 189          # 189 birth records in lbw dataset
low_bw_cases      <- 59          # approximate number with low birth weight
lbw_percent       <- (low_bw_cases / total_births) * 100
lbw_percent       # type the name to print: 31.21693

# More objects with meaningful names
study_name        <- 'Low Birth Weight Study'
```

```
avg_bwt_grams    <- 2945
study_name       # prints: Low Birth Weight Study
```

Key Rule: Object Names in R

- Names are CASE-SENSITIVE: bwt and BWt are different objects
- Names cannot start with a number: 1study is invalid
- Use underscores for spaces: total_births not total births
- Use <- for assignment (= also works but <- is preferred by convention)

Task 1 (Session II)

- Create an object called 'total_infants' with value 200
- Create an object called 'male_infants' with value 102
- Calculate the percentage of male infants and print the result
- What happens if you type Total_Infants instead of total_infants? Why?

Session III — The R Help System

R has a built-in Help system that explains every function. Always check Help before asking Google.

```
# Method 1: Use ? before any function name
?mean           # opens Help page for the mean() function
?sum            # Help for sum()
?table          # Help for frequency tables
?cor.test       # Help for correlation test

# Method 2: Use help()
help('read.csv') # same as ?read.csv

# Method 3: Tab autocomplete
# Type 'me' in the Console and press Tab
# → R suggests: mean(), median(), methods(), merge()...

# Method 4: Find functions by keyword
apropos('test')  # lists all functions with 'test' in the name
```

Reading a Help Page

- Description: what the function does in plain language
- Usage: how to call the function and what arguments it takes
- Arguments: detailed explanation of each parameter
- Value: what the function returns (the output)
- Examples: working code you can copy and paste into your script

Task 2 (Session III)

- Open ?median — what does the argument 'na.rm = TRUE' do?
- Open ?cor.test — what are the possible values for the 'method' argument?
- Type 'me' and press Tab in the Console — list 3 functions that appear

Session IV — Packages & Loading Data

Packages extend R's capabilities. Think of them as apps you install on a phone — install once, use forever.

```
# STEP 1: Install packages (only needs to be done ONCE ever)
# Remove the # to run. After installing, add # back to skip next time.
# install.packages('readxl')      # Read .xlsx and .xls files
# install.packages('tidyverse')   # Data wrangling + ggplot2 graphics
# install.packages('writexl')     # Export data to Excel
# install.packages('broom')       # Clean up regression output

# STEP 2: Load packages — must do this EVERY R session
library(readxl)
library(tidyverse)

# STEP 3: Tell R where your files are saved (working directory)
getwd()    # shows current directory
# In RStudio: Session menu → Set Working Directory → Choose Directory

# STEP 4: Load both datasets
infant <- read_excel('infant.xls')  # loads infant data into object 'infant'
lbw    <- read_excel('lbw.xlsx')    # loads lbw data into object 'lbw'

# STEP 5: First look at the data
head(infant)      # first 6 rows
head(lbw)         # first 6 rows
nrow(infant)      # how many rows? → 200
ncol(infant)      # how many columns? → 9
names(infant)     # variable names
dim(lbw)          # rows × columns together → 189  11
```

Task 3 (Session IV)

- Load both datasets into R using read_excel()
- Use dim() on both datasets — confirm infant is 200×9 and lbw is 189×11
- Use str(lbw) to see variable types — which variable represents birth weight?
- What does the variable 'bwt' contain? Look at its values — what unit might it be in?

DAY 2

Data Structures & Data Wrangling

Types · Factors · Vectors · filter · select · mutate · summarise

Session I: Data Types & Conversion

Session II: Vectors, Factors & Data Frames

Session III: Exploring & Cleaning Data

Session IV: Wrangling with dplyr

Day 2: Data Structures & Data Wrangling

Today you will learn how R stores different kinds of data, and how to clean and transform your datasets using the tidyverse package dplyr.

Session I — Data Types in R

Type	R keyword	Example from our data	How to check
Numeric	numeric	lbw\$bwt = 2551 (birth weight in grams)	class(lbw\$bwt)
Integer	integer	infant\$age = 21 (months)	class(infant\$age)
Character	character	study_name = 'LBW Study'	class(study_name)
Logical	logical	TRUE / FALSE	class(TRUE)
Factor	factor	race_f = White / Black / Other	class(lbw\$race_f)

```
# Check the type of any variable
class(lbw$bwt)      # 'numeric' - birth weight in grams
class(lbw$smoke)    # 'numeric' - but it's really 0/1, needs converting!
class(lbw$race)     # 'numeric' - 1/2/3, also needs labels

# Convert numeric 0/1 to a proper labelled factor
lbw$smoke_factor <- factor(lbw$smoke,
                           levels = c(0, 1),
                           labels = c('Non-Smoker', 'Smoker'))

lbw$race_factor  <- factor(lbw$race,
                           levels = c(1, 2, 3),
                           labels = c('White', 'Black', 'Other'))

lbw$low_factor   <- factor(lbw$low,
                           levels = c(0, 1),
                           labels = c('Normal BW', 'Low BW'))

infant$sex_factor <- factor(infant$sex,
                           levels = c(0, 1),
                           labels = c('Male', 'Female'))

# Now check the frequency table - much more readable!
table(lbw$smoke_factor) # Non-Smoker  Smoker
#               115           74
```

Task 1 (Session I)

- What data type is infant\$sex? Use class() to check
- Convert lbw\$ht (hypertension) to a factor with labels 'No HT' and 'Hypertension'

- Convert `infant$water` to a factor with labels 'No Clean Water' and 'Clean Water'
- How many unique values are in `lbw$race`? Use `length(unique(lbw$race))`

Session II — Vectors, Factors & Data Frames

```
# VECTOR: a list of same-type values
birth_weights <- c(2523, 2551, 2557, 2594, 2600, 1330, 1474)
mean(birth_weights)           # 2232.714
length(birth_weights)         # 7
birth_weights[1]              # first value: 2523
birth_weights[birth_weights < 2500] # filter: 2523 2474 1330 1474

# DATA FRAME: rows = observations, columns = variables
is.data.frame(lbw)            # TRUE — lbw is a data frame
dim(lbw)                      # 189 11 (189 rows, 11 columns)
lbw$bwt                       # access one column with the dollar sign $
lbw[1, ]                      # first row, all columns
lbw[, 'bwt']                  # all rows, 'bwt' column
lbw[lbw$low == 1, ]           # rows where low birth weight = 1
```

Task 2 (Session II)

- Create a vector of the first 10 birth weight values from `lbw$bwt` using `c()`
- Calculate the median of that vector using `median()`
- How many infants in the infant dataset have age > 30 months?
- Use `lbw[lbw$smoke == 1, ]` to extract all smoker rows — how many rows?

Session III — Exploring & Cleaning Data

Before running any statistics, always explore your data first. Know what you have before you analyse it.

```
# Overall summaries
summary(lbw)                  # min, max, mean, median for all variables
str(lbw)                     # structure: types and first few values
glimpse(lbw)                 # tidyverse version of str() — more readable

# Missing values
sum(is.na(lbw))               # total number of missing cells in whole dataset
colSums(is.na(lbw))           # number of missing values per column
sum(is.na(infant))           # check infant too

# Duplicate rows
sum(duplicated(lbw))          # 0 = no duplicates (good!)

# Frequency tables for categorical variables
table(lbw$smoke_factor)       # counts by group
prop.table(table(lbw$low)) * 100 # percentage low birth weight
```

```
table(lbw$smoke_factor, lbw$low_factor)      # cross-tabulation
```

Task 3 (Session III)

- Run `summary(infant)` — what is the minimum value of `wt1`?
- Use `colSums(is.na(lbw))` — are there any missing values in `lbw`?
- What percentage of babies in the `lbw` dataset have low birth weight? Use `prop.table()`
- Create a cross-tabulation of `race_factor` vs `low_factor` in `lbw`

Session IV — Data Wrangling with dplyr

`dplyr` is the core tidyverse package for working with data frames. It uses five key 'verbs':

Verb	What it does	Example
<code>filter()</code>	Keep rows matching a condition	<code>filter(lbw, smoke == 1)</code>
<code>select()</code>	Keep specific columns only	<code>select(lbw, ID, age, bwt)</code>
<code>mutate()</code>	Add or change columns	<code>mutate(lbw, bwt_kg = bwt/1000)</code>
<code>arrange()</code>	Sort rows by a variable	<code>arrange(lbw, bwt)</code>
<code>summarise()</code>	Compute summary statistics	<code>summarise(lbw, Mean=mean(bwt))</code>
<code>group_by()</code>	Split data into groups	<code>group_by(lbw, smoke_factor)</code>

```
library(tidyverse)

# FILTER: keep rows matching a condition
smokers      <- lbw %>% filter(smoke == 1)           # only smokers
young_moms  <- lbw %>% filter(age < 20)             # teenage mothers
smoker_lbwt <- lbw %>% filter(smoke == 1 & low == 1) # smokers WITH low BW
nrow(smokers) # 74 smokers

# SELECT: keep only specific columns
lbw_key      <- lbw %>% select(ID, age, smoke_factor, race_factor, bwt, low)
infant_key   <- infant %>% select(id, age, sex_factor, wt1, ht1, wt2)

# MUTATE: create new columns
lbw <- lbw %>%
  mutate(bwt_kg      = bwt / 1000,
         lbw_category = ifelse(bwt < 2500, 'Low BW', 'Normal BW'),
         age_group    = ifelse(age < 20, 'Teen', 'Adult'))

infant <- infant %>%
  mutate(wt_gain      = wt2 - wt1,
         gained_weight = ifelse(wt_gain > 0, 1, 0))

# ARRANGE: sort rows
lbw %>% arrange(bwt) %>% head(5)      # 5 lightest babies
```

```
lbw %>% arrange(desc(bwt)) %>% head(5) # 5 heaviest babies

# GROUP_BY + SUMMARISE: statistics by group
lbw %>%
  group_by(smoke_factor) %>%
  summarise(Count      = n(),
            Avg_BWT     = round(mean(bwt), 1),
            SD_BWT      = round(sd(bwt), 1),
            LBW_percent = round(mean(low) * 100, 1))
```

Task 4 (Session IV)

- Filter lbw to mothers with hypertension (ht == 1) — how many rows?
 - Create a column 'height_category' in infant: ht1 < 80 = 'Short', else = 'Normal'
 - Use group_by + summarise to find average bwt for each race_factor group
 - Sort infant by wt1 in descending order — which infant has the highest baseline weight?
-

DAY 3

Statistics in Public Health

Descriptive · Normality · t-tests · Chi-square · ANOVA · Correlation · Regression

Session I: Descriptive Statistics

Session II: Normality & Choosing the Right Test

Session III: t-test & Chi-Square

Session IV: ANOVA, Correlation & Regression

Day 3: Statistics in Public Health

Today you will apply statistical tests to answer real public health questions using the infant and lbw datasets. The most important step is choosing the right test for your data.

Session I — Descriptive Statistics

Always describe your data BEFORE running any tests. Ask: what do my variables look like?

```
# Measures of central tendency
mean(lbw$bwt)           # mean birth weight = 2944.665 grams
median(lbw$bwt)         # median = 2977 grams
mean(infant$wt1)        # mean baseline infant weight

# Measures of spread
sd(lbw$bwt)             # standard deviation = 729.02
var(lbw$bwt)            # variance = 531468.8
range(lbw$bwt)          # min and max: 709 4990
IQR(lbw$bwt)            # Q3 - Q1 = interquartile range
quantile(lbw$bwt)       # 0% 25% 50% 75% 100%
summary(lbw$bwt)        # Min Q1 Median Mean Q3 Max all together

# Grouped descriptive stats (most useful in public health!)
lbw %>%
  group_by(smoke_factor) %>%
  summarise(n           = n(),
            Mean_BWT    = round(mean(bwt), 1),
            Median_BWT  = median(bwt),
            SD_BWT      = round(sd(bwt), 1),
            LBW_pct     = round(mean(low)*100, 1))
#   smoke_factor    n   Mean_BWT Median_BWT SD_BWT LBW_pct
#   <fct>          <dbl> <dbl>      <dbl>   <dbl> <dbl>
# 1 Non-Smoker    115  3055.7    3100     752.4  28.7
# 2 Smoker        74  2771.9    2775     659.8  40.5
```

Task 1 (Session I)

- Calculate mean, median, and SD of infant\$wt1 (baseline height)
- What is the IQR of lbw\$lwt (last weight before delivery)?
- Make a grouped summary of infant data by sex_factor — include avg wt1, wt2, and wt_gain
- What percentage of infants had access to clean water (water == 1)?

Session II — Normality Testing & Choosing the Right Test

Before choosing a statistical test, you must check whether your continuous outcome variable follows a normal (bell-shaped) distribution.

Situation	Data Normal?	Parametric Test	Non-Parametric Alternative
Compare 2 independent groups	Yes	t.test()	wilcox.test()
Compare same group (before/after)	Yes	t.test(paired=TRUE)	wilcox.test(paired=TRUE)
Compare 3+ groups	Yes	aov()	kruskal.test()
Two categorical variables	N/A	chisq.test()	fisher.test() (small n)
Linear relationship	Yes	cor.test(pearson)	cor.test(spearman)

```
# Shapiro-Wilk test for normality
# H0: the data IS normally distributed
# Decision rule:
#   p >= 0.05 → Cannot reject H0 → assume normal → use parametric test
#   p < 0.05 → Reject H0 → NOT normal → use non-parametric test

shapiro.test(lbw$bwt)          # birth weight
# W = 0.97843, p-value = 0.02001 → NOT normal at 0.05 level

shapiro.test(infant$wt1)       # infant baseline weight
shapiro.test(infant$wt_gain)   # weight gain

# Visual check — quick histogram
hist(lbw$bwt, main='Birth Weight Distribution', xlab='Grams')
```

Task 2 (Session II)

- Run shapiro.test on lbw\$sage (mother's age) — is it normally distributed?
- Run shapiro.test on infant\$ht1 (infant height) — is it normally distributed?
- Based on your results, which test would you use to compare height by sex?

Session III — Hypothesis Testing: t-test & Chi-Square

The Hypothesis Testing Framework

- H0 (Null hypothesis): there is NO difference / association
- H1 (Alternative hypothesis): there IS a difference / association
- $p < 0.05$: reject H0 → result is statistically significant
- $p \geq 0.05$: do not reject H0 → result is not significant
- ALWAYS state your H0 and H1 BEFORE looking at results

```
# — INDEPENDENT t-TEST —
# Q: Is birth weight different between smokers and non-smokers?
```



```
# H0: No difference in mean BWT | H1: There IS a difference
t.test(bwt ~ smoke_factor, data = lbw)
# t = 2.7299, df = 170.1, p-value = 0.007003
# mean in Non-Smoker: 3055.696 | mean in Smoker: 2771.919
# → p < 0.05, significant difference

# — PAIRED t-TEST —
# Q: Did infant weight change significantly from wt1 to wt2?
# Same infants measured twice → must use paired = TRUE
t.test(infant$wt1, infant$wt2, paired = TRUE)
# If p < 0.05 → significant weight gain

# — WILCOXON TEST (non-parametric alternative to t-test) —
wilcox.test(bwt ~ smoke_factor, data = lbw)
wilcox.test(infant$wt1, infant$wt2, paired = TRUE)

# — CHI-SQUARE TEST —
# Q: Is smoking associated with low birth weight?
# Both variables are categorical → chi-square
chi_tab <- table(lbw$smoke_factor, lbw$low_factor)
chi_tab
#           Normal BW  Low BW
# Non-Smoker       82     33
# Smoker           44     30
chisq.test(chi_tab)
# X-squared = 4.9237, df = 1, p-value = 0.02649 → significant
```

Task 3 (Session III)

- Is birth weight (bwt) significantly different between hypertensive and non-hypertensive mothers? Run `t.test(bwt ~ ht, data=lbw)`
- Is race associated with low birth weight? Run `chisq.test(table(lbw$race_factor, lbw$low_factor))`
- Did infant height (ht1) differ significantly by sex? State your H0 first, then run the test
- Run a paired t.test on infant wt1 vs wt2 — what is the mean difference?

Session IV — ANOVA, Correlation & Linear Regression

```
# — ONE-WAY ANOVA —
# Q: Does birth weight differ across racial groups?
# H0: All group means are equal
anova_result <- aov(bwt ~ race_factor, data = lbw)
summary(anova_result)
#           Df    Sum Sq Mean Sq F value Pr(>F)
# race_factor 2  5015725 2507862   4.913  0.00834 **
# → p < 0.05: at least one group is different

# Post-hoc test (which groups differ?)
TukeyHSD(anova_result)
```

```
# ——— PEARSON CORRELATION ———
# Q: Is mother's weight (lwt) correlated with baby's weight (bwt)?
cor.test(lbw$lwt, lbw$bwt, method = 'pearson')
# cor = 0.18, p-value = 0.013 → weak positive correlation, significant

# Infant: height vs weight at baseline
cor.test(infant$htl, infant$wtl, method = 'pearson')

# ——— SIMPLE LINEAR REGRESSION ———
# Q: Does mother's age predict birth weight?
model <- lm(bwt ~ age, data = lbw)
summary(model)
# Estimate for age: ~5 grams per year (p = 0.31, not significant)
# R-squared: 0.0056 (explains only 0.56% of variation)

# ——— MULTIPLE LINEAR REGRESSION ———
model_multi <- lm(bwt ~ age + lwt + smoke, data = lbw)
summary(model_multi)
# smoke coefficient: -278 (smokers' babies weigh ~278g less, p<0.05)
```

Task 4 (Session IV)

- ANOVA: Does infant follow-up weight (wt2) differ by mother's education level (maeduc)?
- Correlation: What is the Pearson correlation between infant age and wt1?
- Regression: Build `lm(wt2 ~ wt1, data=infant)` — what is the R-squared? Interpret it.
- CHALLENGE: Add age as a second predictor to the model above. Did R-squared improve?

DAY 4

Data Visualization with ggplot2

Histogram · Bar Chart · Boxplot · Scatter · Facets · Themes · Saving

Session I: Grammar of Graphics

Session II: Basic Plot Types

Session III: Customizing Plots

Session IV: Advanced Plots & Saving

Day 4: Data Visualization with ggplot2

ggplot2 is the most powerful visualization package in R. It builds plots in layers using a consistent 'grammar of graphics' — you add elements one at a time with the + sign.

Session I — The Grammar of Graphics

Layer	Code	What it does
Canvas	<code>ggplot(data, aes(x=, y=))</code>	Defines which dataset and which variables map to axes
Geometry	<code>geom_histogram()</code>	Defines the type of plot (histogram, bar, box, point...)
Labels	<code>labs(title=, x=, y=)</code>	Adds title and axis labels to make plot self-explanatory
Theme	<code>theme_classic() / theme_bw()</code>	Controls the overall look: background, gridlines, fonts

```
library(tidyverse)  # loads ggplot2 automatically

# The simplest plot: one line
ggplot(lbw, aes(x = bwt)) + geom_histogram()

# Add layers one by one with +
ggplot(lbw, aes(x = bwt)) +
  geom_histogram(fill = 'steelblue', color = 'white', bins = 20) +
  labs(title = 'Distribution of Birth Weight',
        x     = 'Birth Weight (grams)',
        y     = 'Number of Babies') +
  theme_classic()
```

Task 1 (Session I)

- Create a histogram of `infant$wt1` (baseline weight)
- Make the bars green (`fill='green'`) with white borders (`color='white'`)
- Add a title and proper axis labels using `labs()`
- Try `theme_bw()` instead of `theme_classic()` — describe the difference

Session II — Basic Plot Types

```
# 1. HISTOGRAM — distribution of one continuous variable
ggplot(lbw, aes(x = bwt)) +
  geom_histogram(binwidth = 250, fill = 'coral', color = 'white') +
  labs(title = 'Birth Weight Distribution', x = 'Grams', y = 'Count') +
  theme_minimal()

# 2. BAR CHART — count per category
ggplot(lbw, aes(x = race_factor)) +
```

```

geom_bar(fill = 'steelblue') +
labs(title = 'Race Distribution', x = 'Race', y = 'Count') +
theme_classic()

# 3. BOXPLOT — compare groups on a continuous outcome
ggplot(lbw, aes(x = smoke_factor, y = bwt, fill = smoke_factor)) +
  geom_boxplot() +
  labs(title = 'Birth Weight by Smoking Status',
        x = 'Smoking Status', y = 'Birth Weight (grams)') +
  theme_classic()

# 4. SCATTER PLOT — relationship between two continuous variables
ggplot(lbw, aes(x = lwt, y = bwt)) +
  geom_point(alpha = 0.5, color = 'darkblue') +
  labs(title = 'Mother Weight vs Birth Weight',
        x = 'Mother Last Weight (lbs)', y = 'Birth Weight (grams)') +
  theme_classic()

```

Task 2 (Session II)

- Create a bar chart showing count of low_factor (Normal BW vs Low BW)
- Make a boxplot comparing infant wt1 by sex_factor — use fill=sex_factor
- Create a scatter plot: infant age (x) vs wt1 (y) — set alpha=0.6 and color='tomato'
- What does 'alpha' control? Try alpha=0.1 and alpha=1.0 to see the difference

Session III — Customizing Plots

```

# Grouped bar chart (two categorical variables side by side)
ggplot(lbw, aes(x = race_factor, fill = low_factor)) +
  geom_bar(position = 'dodge') +      # 'dodge'=side-by-side; 'fill'=proportions
  labs(title = 'Low Birth Weight by Race', fill = 'BW Status') +
  scale_fill_manual(values = c('Normal BW'='#2196F3', 'Low BW'='#F44336')) +
  theme_classic()

# Boxplot with individual data points (best practice!)
ggplot(lbw, aes(x = smoke_factor, y = bwt, fill = smoke_factor)) +
  geom_boxplot(alpha = 0.7, outlier.shape = NA) +
  geom_jitter(width = 0.2, alpha = 0.3, size = 1.5) +
  labs(title = 'Birth Weight by Smoking Status',
        subtitle = 'Boxes: median ± IQR; dots: individual births',
        x = 'Smoking Status', y = 'Birth Weight (grams)') +
  scale_fill_manual(values = c('Non-Smoker'='#4CAF50', 'Smoker'='#E91E63')) +
  theme_classic() +
  guides(fill = 'none')      # remove legend (info already on x-axis)

# Scatter with regression line, colored by group
ggplot(lbw, aes(x = age, y = bwt, color = smoke_factor)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = 'lm', se = TRUE) +
  theme_classic()

```

Task 3 (Session III)

- Create a grouped bar chart: `x=water_factor`, `fill=sex_factor` (infant data), `position='fill'`
- Make a boxplot of infant `wt_gain` by `water_factor` and add `geom_jitter()`
- Add a regression line to a scatter plot of infant age vs `wt2`, color points by `sex_factor`
- Try `theme_bw()` and add `theme(plot.title = element_text(face='bold'))` to make the title bold

Session IV — Facets, Density Plots & Saving

```
# FACETING: one panel per group
ggplot(lbw, aes(x = age, y = bwt, color = low_factor)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = 'lm', se = FALSE) +
  facet_wrap(~ race_factor) + # one panel per race group
  labs(title = 'Age vs Birth Weight by Race', color = 'BW Status') +
  theme_bw()

# DENSITY PLOT: smooth distribution curve
ggplot(lbw, aes(x = bwt, fill = smoke_factor)) +
  geom_density(alpha = 0.4) +
  labs(title = 'Birth Weight Density by Smoking Status') +
  theme_classic()

# VIOLIN PLOT: shows distribution shape + boxplot inside
ggplot(lbw, aes(x = sex_factor, y = wt1, fill = sex_factor)) +
  geom_violin(alpha = 0.6) +
  geom_boxplot(width = 0.1, fill = 'white') +
  theme_classic() + guides(fill = 'none')

# SAVING PLOTS with ggsave()
my_plot <- ggplot(lbw, aes(x = smoke_factor, y = bwt, fill = smoke_factor)) +
  geom_boxplot() + theme_classic()

ggsave('birthweight_smoking.png', # filename
       plot = my_plot,           # which plot object
       dpi = 300,                # resolution (300 = print quality)
       width = 8,                # width in inches
       height = 6)               # height in inches

ggsave('birthweight_smoking.pdf', plot = my_plot, width = 8, height = 6)
```

Task 4 (Session IV)

- Create a faceted density plot of `lbw$bwt`, faceted by `race_factor`, filled by `smoke_factor`
- Build a stacked proportional bar chart for infant: `x=sex_factor`, `fill=water_factor`, `position='fill'`
- Save your best plot as a PNG with `dpi=300`, `width=10`, `height=7`

- CHALLENGE: Make a scatter plot with 3 variables: age vs bwt, faceted by race_factor, colored by smoke_factor

DAY 5

Logistic Regression, Reporting & Group Project

`glm()` · Odds Ratios · Forest Plots · Summary Tables · Mini-Project

Session I: Logistic Regression with `glm()`

Session II: Summary Tables & Forest Plots

Session III: Complete Analysis Workflow

Session IV: Group Work Mini-Project

Day 5: Logistic Regression, Reporting & Group Project

Today you will learn how to analyse binary (yes/no) outcomes using logistic regression — the most common method in public health research. You will also learn to produce clean tables and visualisations for reports, and apply everything in a group project.

Session I — Logistic Regression with glm()

When to use Logistic Regression

- Your OUTCOME variable is binary: low birth weight (yes/no), disease (yes/no), death (yes/no)
- You want to identify risk factors and control for confounders simultaneously
- Results are expressed as Odds Ratios (OR) with 95% Confidence Intervals
- $OR > 1$ = risk factor (increases odds of outcome)
- $OR < 1$ = protective factor (decreases odds of outcome)
- $OR = 1$ = no association with the outcome

```
library(broom)    # for tidy model output

# ——— SIMPLE LOGISTIC REGRESSION ———
# Q: Does maternal smoking predict low birth weight?
# Outcome (Y): low (0=normal, 1=low birth weight)
# Predictor (X): smoke (0/1)
modell1 <- glm(low ~ smoke, data = lbw, family = binomial)
summary(modell1)

# Convert log-odds to Odds Ratios (ALWAYS do this!)
exp(coef(modell1))      # Odds Ratios: (Intercept)=0.40, smoke=2.02
exp(confint(modell1))   # 95% Confidence Intervals
# smoke OR = 2.02: smokers have ~2× higher odds of having a low-BW baby

# Tidy formatted output using broom package
tidy(modell1, exponentiate = TRUE, conf.int = TRUE)
#   term          estimate conf.low conf.high p.value
# (Intercept)    0.40      0.25      0.64 <0.001
# smoke          2.02      1.08      3.78  0.028

# ——— MULTIPLE LOGISTIC REGRESSION ———
# Q: What factors independently predict low birth weight?
modell2 <- glm(low ~ age + smoke + race + ht + ui + lwt,
              data = lbw, family = binomial)
summary(modell2)

# Odds ratio table for all predictors
tidy(modell2, exponentiate = TRUE, conf.int = TRUE)

# Compare model fit: AIC (lower = better fit)
AIC(modell1)    # simple model
```

```
AIC(model2)    # full model

# ——— LOGISTIC REGRESSION FOR INFANT DATA ———
# Q: Does clean water access predict infant weight gain?
model3 <- glm(gained_weight ~ water + sex + age + wtl,
              data = infant, family = binomial)
tidy(model3, exponentiate = TRUE, conf.int = TRUE)
```

Task 1 (Session I)

- Run simple logistic regression: `low ~ ht` (hypertension). What is the OR? Is it significant?
- Add race to the model: `low ~ ht + race`. Does the OR for hypertension change? What does this suggest?
- In the infant dataset, run: `gained_weight ~ sex + age`. Which predictor is significant?
- Compare AIC of model with and without age. Which model fits better?

Session II — Summary Tables & Forest Plots

```
# ——— DESCRIPTIVE SUMMARY TABLE ———
lbw %>%
  group_by(smoke_factor) %>%
  summarise(n          = n(),
            Mean_Age    = round(mean(age), 1),
            Mean_BWT_g  = round(mean(bwt), 0),
            LBW_pct     = round(mean(low)*100, 1),
            HT_pct      = round(mean(ht)*100, 1))

# ——— FORMATTED OR TABLE ———
or_table <- tidy(model2, exponentiate = TRUE, conf.int = TRUE) %>%
  filter(term != '(Intercept)') %>%
  mutate(OR = round(estimate, 2),
         CI = paste0('(', round(conf.low,2), ', ', round(conf.high,2), ')'),
         p  = round(p.value, 3),
         Sig = ifelse(p.value < 0.05, 'Yes *', 'No')) %>%
  select(Predictor = term, OR, '95% CI' = CI, 'p-value' = p, Significant = Sig)
print(or_table)

# Save results table to CSV (open in Excel/Word)
write.csv(or_table, 'logistic_results.csv', row.names = FALSE)

# ——— FOREST PLOT ———
# Forest plots display odds ratios visually with confidence intervals
or_data <- tidy(model2, exponentiate = TRUE, conf.int = TRUE) %>%
  filter(term != '(Intercept)')

ggplot(or_data, aes(x = estimate, y = reorder(term, estimate))) +
  geom_point(size = 3, color = 'darkblue') +
  geom_errorbarh(aes(xmin = conf.low, xmax = conf.high), height = 0.25) +
```

```
geom_vline(xintercept = 1, linetype = 'dashed', color = 'red') +
labs(title = 'Predictors of Low Birth Weight: Odds Ratios',
      subtitle = 'Dashed line = OR of 1 (no effect)',
      x = 'Odds Ratio (95% CI)',
      y = 'Predictor Variable') +
theme_classic()

ggsave('forest_plot.png', dpi = 300, width = 8, height = 5)
```

Task 2 (Session II)

- Create a descriptive summary table for the infant dataset: n, mean age, mean wt1, mean wt2, mean wt_gain, % male, % clean water
- Using model3 (infant logistic regression), create a formatted OR table
- Save the OR table as 'infant_logistic_results.csv'
- Build a forest plot for model3 and save as 'infant_forest_plot.png' with dpi=300

Session III — Complete Analysis Workflow

This session demonstrates a complete analysis from raw data to publication-ready output — the full pipeline you will use in your own research.

```
# == STEP 1: Load and prepare data ==
library(readxl); library(tidyverse); library(broom)

lbw_analysis <- read_excel('lbw.xlsx') %>%
  mutate(smoke_f = factor(smoke, levels=c(0,1),
                          labels=c('Non-Smoker', 'Smoker')),
         race_f = factor(race, levels=c(1,2,3),
                          labels=c('White', 'Black', 'Other')),
         low_f = factor(low, levels=c(0,1),
                          labels=c('Normal', 'Low BW'))

# == STEP 2: Descriptive summary by outcome ==
lbw_analysis %>%
  group_by(low_f) %>%
  summarise(n=n(), Mean_Age=mean(age), Smoke_pct=mean(smoke)*100,
            HT_pct=mean(ht)*100, Mean_BWT=mean(bwt))

# == STEP 3: Statistical test ==
chisq.test(table(lbw_analysis$smoke_f, lbw_analysis$low_f))

# == STEP 4: Logistic regression ==
final_model <- glm(low ~ age + smoke + race + ht + lwt,
                  data=lbw_analysis, family=binomial)

# == STEP 5: Forest plot ==
tidy(final_model, exponentiate=TRUE, conf.int=TRUE) %>%
  filter(term != '(Intercept)') %>%
```

```
ggplot(aes(x=estimate, y=reorder(term, estimate))) +
  geom_point(size=3, color='darkblue') +
  geom_errorbarh(aes(xmin=conf.low, xmax=conf.high), height=0.25) +
  geom_vline(xintercept=1, linetype='dashed', color='red') +
  theme_classic() +
  labs(title='Predictors of Low Birth Weight', x='OR (95% CI)', y='')
```

```
# == STEP 6: Save outputs ==
ggsave('final_forest_plot.png', dpi=300, width=8, height=5)
```

Task 3 (Session III)

- Apply the complete workflow to the infant dataset
- Outcome variable: gained_weight (did the infant gain weight: 0/1)
- Step 3: Run chi-square for water access vs weight gain
- Steps 4–6: Run logistic regression, build forest plot, save as PNG

Session IV — Group Work Mini-Project

This is your opportunity to apply everything you have learned over 5 days. Work in groups of 3–4 participants and present your findings to the class.

Group Work Instructions

- Groups of 3–4 participants per group
- Choose ONE dataset: infant OR lbw
- Write ONE clear research question before you start
- Complete all 6 steps below using R
- Prepare a 10-minute presentation for the class
- Use the checklist at the end to review your work

Step	What to do	R Code to Use
1	Load and prepare data — add all factor labels	<code>read_excel()</code> , <code>factor()</code> , <code>mutate()</code>
2	Descriptive statistics by key groups	<code>group_by()</code> , <code>summarise()</code>
3	At least 2 meaningful visualisations	<code>ggplot()</code> , <code>geom_boxplot()</code> , <code>geom_bar()</code>
4	Statistical test — state H0 and H1 first	<code>chisq.test()</code> or <code>t.test()</code> or <code>aov()</code>
5	Logistic or linear regression with table	<code>glm()</code> or <code>lm()</code> , <code>tidy()</code> , <code>exp(coef())</code>
6	Forest plot or final polished visualisation	<code>ggplot()</code> + <code>geom_point()</code> + <code>ggsave()</code>

Presentation Checklist

- Research question stated clearly in one sentence
- Dataset described: n, key variables, any cleaning done
- At least 2 plots shown and explained
- Statistical test result reported: test statistic + p-value + interpretation
- Regression table with OR (or Beta coefficients) and 95% CI
- One clear conclusion sentence answering the research question

Quick Reference: Common R Functions

Data Loading & Exploration

Function	Purpose	Example
<code>read_excel()</code>	Load Excel file (.xlsx or .xls)	<code>infant <- read_excel('infant.xls')</code>
<code>head()</code>	Show first 6 rows	<code>head(lbw)</code>
<code>tail()</code>	Show last 6 rows	<code>tail(lbw)</code>
<code>dim()</code>	Rows and columns	<code>dim(infant) # 200 9</code>
<code>names()</code>	Column names	<code>names(lbw)</code>
<code>str()</code>	Variable types and preview	<code>str(lbw)</code>
<code>summary()</code>	Summary statistics for all variables	<code>summary(infant)</code>
<code>glimpse()</code>	Tidyverse version of <code>str()</code>	<code>glimpse(lbw)</code>
<code>class()</code>	Data type of a variable	<code>class(lbw\$bwt) # 'numeric'</code>
<code>is.na()</code>	Check for missing values	<code>sum(is.na(lbw))</code>

Data Wrangling (dplyr)

Function	Purpose	Example
<code>filter()</code>	Keep rows meeting condition	<code>lbw %>% filter(smoke==1)</code>
<code>select()</code>	Keep specific columns	<code>lbw %>% select(ID, bwt, low)</code>
<code>mutate()</code>	Add or change columns	<code>lbw %>% mutate(bwt_kg=bwt/1000)</code>
<code>arrange()</code>	Sort by variable	<code>lbw %>% arrange(desc(bwt))</code>
<code>group_by()</code>	Group by a variable	<code>lbw %>% group_by(smoke_factor)</code>
<code>summarise()</code>	Summarize data	<code>summarise(n=n(), Mean=mean(bwt))</code>
<code>factor()</code>	Convert to factor with labels	<code>factor(x, levels=c(0,1), labels=c('No','Yes'))</code>
<code>ifelse()</code>	Conditional new values	<code>ifelse(bwt<2500,'Low BW','Normal')</code>

Statistical Tests

Function	When to use	Example
<code>t.test()</code>	Compare means: 2 independent groups	<code>t.test(bwt ~ smoke_f, data=lbw)</code>
<code>t.test(paired)</code>	Compare means: paired (before/after)	<code>t.test(wt1, wt2, paired=TRUE)</code>

Function	When to use	Example
wilcox.test()	Non-parametric t-test alternative	wilcox.test(bwt ~ smoke_f, data=lbw)
chisq.test()	Association between 2 categories	chisq.test(table(smoke_f, low_f))
aov()	Compare means: 3+ groups (ANOVA)	aov(bwt ~ race_f, data=lbw)
TukeyHSD()	Post-hoc: which ANOVA groups differ?	TukeyHSD(anova_result)
kruskal.test()	Non-parametric ANOVA	kruskal.test(bwt ~ race_f, data=lbw)
shapiro.test()	Test for normality	shapiro.test(lbw\$bwt)
cor.test()	Correlation between two variables	cor.test(lwt, bwt, method='pearson')
lm()	Linear regression	lm(bwt ~ age + smoke, data=lbw)
glm(family=binomial)	Logistic regression	glm(low ~ smoke+race, family=binomial)

ggplot2 Quick Reference

Geom / Function	What it creates	Key Arguments
geom_histogram()	Distribution of continuous variable	bins=, binwidth=, fill=, color=
geom_bar()	Count per category	fill=, position='dodge'/'fill'
geom_boxplot()	Median ± IQR by group	fill=, alpha=, outlier.shape=
geom_violin()	Distribution shape + boxplot	alpha=, fill=
geom_point()	Scatter plot	alpha=, color=, size=
geom_smooth()	Regression / trend line	method='lm', se=TRUE/FALSE
geom_jitter()	Show individual points on boxplot	width=, alpha=, size=
geom_density()	Smooth density curve	alpha=, fill=
facet_wrap(~var)	Multiple panels by group	ncol=, nrow=
labs()	Title, axes, legend labels	title=, x=, y=, fill=, color=
theme_classic()	Clean white background no grid	-
ggsave()	Save plot to file	dpi=, width=, height=

Next Steps After This Workshop

Practice on Your Own Data

- Repeat the Day 2–5 workflow on your own dataset as soon as possible
- Try to answer one research question from your own work using R
- Build one visualization that you would be proud to show in a report

Recommended Learning Resources

Resource	What it covers	Access
R for Data Science (Hadley Wickham)	Tidyverse, ggplot2, R Markdown — the bible of R	Free: r4ds.had.co.nz
Introduction to R (swirl package)	Interactive in-console R tutorials	<code>install.packages('swirl')</code>
Stack Overflow	Answer to almost every R question ever asked	stackoverflow.com
R-Bloggers	Daily tutorials and applied examples in R	r-bloggers.com
UCLA IDRE Stats	Statistical methods with R examples	stats.oarc.ucla.edu/r/
Posit Community	Official RStudio/Posit community forum	community.posit.co

Useful Packages to Explore Next

Package	Purpose
gtsummary	Automatically creates beautiful Table 1 (descriptive statistics) for papers
ggpubr	Add p-values and significance brackets directly to ggplot2 figures
rmarkdown	Create reproducible HTML/Word/PDF reports mixing code and text
shiny	Build interactive web apps with R — no HTML/CSS knowledge needed
lubridate	Work with dates and times in R easily
tableone	Create Table 1 for clinical research papers quickly

Final Message from the Facilitators

Thank you for your participation and enthusiasm throughout this training. R is a tool that rewards practice — the more you use it, the more natural it becomes. Keep your R scripts from this workshop, revisit them when you are stuck, and remember: every expert started with hello world. We hope this workshop has given you the foundation to bring data-driven insights into your public health work. Good luck!

— End of Workshop Handout —
